

REnFormer : A Domain-Specific Transformer for Multi-Site Solar Power Forecasting

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Abstract

Time-series foundation models (TSFMs) promise accurate zero-shot forecasting from a single large pre-trained network, and recent studies report that they outperform classical baselines even in specialised domains. We ask whether this scale is necessary when abundant domain data is available. We present **REnFormer**, a compact domain-specific Transformer (536 k parameters) for probabilistic multi-site solar power forecasting, trained on 291 photovoltaic plants from Chile’s national electricity system (SEN) covering 5.4 years of hourly generation. REnFormer combines per-window reversible instance normalisation with a shared encoder, a macro skip connection, and a Gaussian output head trained with a *masked negative log-likelihood* that concentrates learning on non-trivial generation steps. The proposed model is compared against TimesFM 2.5 (231M parameters, zero-shot) over fleet sizes from 10 to 291 sites, REnFormer matches the foundation model on MAE at full fleet scale (1.936 vs. 1.936 MW) and improves RMSE by 3.2% and CRPS by 5.1%, while using $431\times$ fewer parameters and running $\sim 640\times$ faster per forecast (0.036 ms vs. 23.0 ms). The probabilistic advantage holds at *every* fleet size evaluated. Our results add domain-specific evidence to a growing literature suggesting that small, specialised forecasting models remain a competitive choice when enough in-domain data exists, relegating TSFMs to cold-start and data-scarce regimes.

1 Introduction

Time-series forecasting is undergoing the same scale-driven transition that reshaped natural language processing: large pre-trained *time-series foundation models* (TSFMs) such as TimesFM [1] offer zero-shot forecasts from a single network with hundreds of millions of parameters, and LLM-based forecasters repurpose

billion-parameter language backbones for numeric sequences [2–4]. Recent evaluations report that TSFMs beat classical econometric baselines out of the box, even in specialised domains such as macroeconomic forecasting [5].

A growing counter-literature, however, questions whether this scale is doing the work. Ablation studies show that removing the pre-trained LLM from LLM-based forecasters often does not hurt—and sometimes improves—accuracy [6]. Sub-3B “small language model” forecasters match or exceed 7B baselines at a fraction of the training cost [7]; compact architectures distil the inductive biases of large vision-based forecasters into models three orders of magnitude smaller [8]; and recent lightweight foundation models (23M–100M parameters) trained on modest hardware reach state-of-the-art benchmark performance [9, 10]. The emerging picture is that for time series—unlike text—parameter count is a poor proxy for forecasting skill, and that small, well-designed models are competitive whenever relevant training data exists.

This paper provides a controlled, operationally grounded data point for that debate in the renewable energy domain. Solar power forecasting at grid scale is an ideal test bed: national system operators hold years of hourly generation history for every plant (so the data-scarcity argument for TSFMs does not apply), the fleet is large and heterogeneous (so per-site models do not scale), and forecasts must be produced for hundreds of sites every hour (so inference cost matters). Solar generation is also structurally hard for generic pre-trained models: it is exactly zero for roughly half of all timesteps and highly intermittent otherwise, a regime that generic training corpora under-represent.

We present **REnFormer** (**R**enewable **E**nergy **T**rans**F**ormer), a domain-specific Transformer with 536 k parameters—0.23% the size of TimesFM 2.5—trained globally on 291 solar plants of Chile’s national electricity system (SEN). Our contributions are:

1. A **compact global architecture**: a shared 4-layer encoder with per-window reversible instance normalisation (the same input scaling principle TSFMs rely on), a macro skip connection, and a probabilistic Gaussian head, totalling 536 k parameters.
2. A **masked Gaussian NLL objective** that focuses learning on non-trivial generation steps (> 0.1 MW), directly addressing the intermittency structure that generic foundation models must handle implicitly.
3. A **direct comparison against TimesFM 2.5 zero-shot** on identical test windows across fleet sizes from 10 to 291 sites. At full fleet scale REnFormer matches the foundation model on MAE and improves RMSE (3.2%) and CRPS (5.1%), at $431\times$ fewer parameters and $\sim 640\times$ lower inference latency; its probabilistic advantage holds at every fleet size.

2 Related Work

2.1 Foundation Models and LLMs for Time Series

TimesFM [1] established the decoder-only pre-trained foundation-model recipe for forecasting; a parallel line adapts language models directly, surveyed in [2]. AutoTimes [3] repurposes frozen decoder-only LLMs autoregressively with only 0.1% trainable parameters, and LangTime [4] adds cross-domain pre-training and reinforcement-learning fine-tuning to stabilise autoregressive rollout. Foundation models can also be adapted cheaply at inference time: TFMAdapter [11] injects covariates into frozen TSFM forecasts with a lightweight instance-level adapter, gaining 24–27% over the base models. In application studies, zero-shot TSFMs (TimesFM, Moirai) outperformed econometric baselines for macroeconomic forecasting across 27 EU countries [5]—evidence that motivates our choice of TimesFM 2.5 as the strongest available zero-shot comparator.

2.2 Small and Lightweight Forecasting Models

The necessity of scale is increasingly contested. Tan et al. [6] show through ablation that the pre-trained LLM component of several prominent LLM-forecasters can be removed without loss of accuracy, attributing reported gains to the surrounding architecture. SME-Times [7] demonstrates that sub-3B models with statistically enhanced prompts and a mixture-of-experts

outperform 7B baselines with far less compute. SV-Time [8] encodes the inductive biases of large vision-model forecasters into a small model with $10^3\times$ fewer parameters, and Tiny-TSM [9] trains a 23M-parameter foundation model to state-of-the-art medium/long-horizon accuracy on a single GPU. SEMPO [10] reaches strong zero- and few-shot performance with a small footprint via spectral decomposition and prompt mixtures. These works compress or replace *general-purpose* models; REnFormer is complementary, asking how far a small model goes when it is *specialised*: trained from scratch on a single domain with abundant history, and equipped with domain-aware components (intermittency masking, per-window renormalisation).

2.3 Multi-Site PV Forecasting

Global models—one parameterisation shared across all series [12]—have gained traction for PV fleets: Jang et al. [13] adapt a shared model to known and unseen Korean sites, and Dimitriadis et al. [14] scale to three interconnected European countries using input normalisation against cross-border distribution shift. Transformer architectures dominate recent work: the Temporal Fusion Transformer for day-ahead PV [15], inter-series attention for sparse demand [16], and Transformers for intermittent EV load [17]. Spatial-temporal GNNs [18–20] model inter-site dependencies explicitly but require a pre-defined adjacency structure and $\mathcal{O}(N^2)$ edge computation. REnFormer operates in the purely global regime—no per-site parameters, no graph—which is what allows a single 536 k-parameter network to serve an entire national fleet.

3 Problem Formulation

Let $\mathcal{S} = \{1, \dots, N\}$ denote a fleet of $N = 291$ solar plants. For each plant $s \in \mathcal{S}$ and time t , let $y_{s,t} \in \mathbb{R}_{\geq 0}$ denote the hourly generation in MW.

Global multi-site forecasting. Given a lookback window of T consecutive hours, $\mathbf{x}_s = (y_{s,t-T+1}, \dots, y_{s,t}) \in \mathbb{R}^T$, predict the next H hourly values $\mathbf{y}_s = (y_{s,t+1}, \dots, y_{s,t+H}) \in \mathbb{R}^H$ using a single shared model f_θ such that:

$$\hat{\mathbf{y}}_s = f_\theta(\mathbf{x}_s), \quad \forall s \in \mathcal{S}. \quad (1)$$

The model parameters θ are shared across all sites; sites contribute independently as training instances, exactly as series do for a zero-shot foundation model. This makes the comparison with TimesFM 2.5 like-for-like: both models map a raw-MW history window to a forecast, relying on internal per-window normalisation

to handle the three-orders-of-magnitude spread in plant capacity.

Intermittency mask. Define a binary mask $\mathbf{m}_s \in \{0, 1\}^H$ with $m_{s,t+h} = \mathbf{1}[y_{s,t+h} > \epsilon]$ for a threshold $\epsilon > 0$ (we set $\epsilon = 0.1$ MW). Masked timesteps correspond to trivially near-zero generation steps (night-time, zero-irradiance) where prediction accuracy is operationally irrelevant.

Probabilistic objective. f_θ outputs Gaussian parameters $(\boldsymbol{\mu}_s, \boldsymbol{\sigma}_s^2) \in \mathbb{R}^H \times \mathbb{R}_{>0}^H$. The training objective is the masked negative log-likelihood (NLL):

$$\mathcal{L} = -\frac{1}{|\mathcal{B}|} \sum_{s \in \mathcal{B}} \frac{\sum_{h=1}^H m_{s,t+h} \log \mathcal{N}(y_{s,t+h}; \boldsymbol{\mu}_{s,h}, \boldsymbol{\sigma}_{s,h}^2)}{\sum_{h=1}^H m_{s,t+h} + \delta}, \quad (2)$$

where $\mathcal{B}$ is a mini-batch and $\delta = 10^{-6}$ prevents division by zero on all-dark batches.

4 REnFormer Architecture

REnFormer is deliberately small: the design question is not how much capacity can be added, but which components a compact model needs to absorb a heterogeneous national fleet.

4.1 Per-Window Reversible Normalisation

The model receives *raw MW* input. Each lookback window’s power channel is normalised by its own statistics before encoding, and the Gaussian outputs are mapped back to MW by the inverse transform—the reversible instance normalisation (RevIN) principle [21], and the same mechanism that lets TimesFM ingest unscaled series. This preserves per-site capacity information in the window statistics while letting a single encoder serve plants from < 1 MW rooftops to ~ 390 MW utility installations. The base variant uses RevIN with learnable affine parameters; the skip variant (below) uses plain per-window mean/std normalisation. Exogenous calendar channels (sin/cos hour and day-of-year) bypass normalisation.

4.2 Input Encoding

A learned linear projection maps each timestep from $\mathbb{R}^F$ to $\mathbb{R}^d$ (F input channels, model dimension d):

$$\mathbf{h}_t^{(0)} = \mathbf{W}_{\text{in}} \mathbf{x}_{s,t} + \mathbf{b}_{\text{in}}, \quad \mathbf{W}_{\text{in}} \in \mathbb{R}^{d \times F}. \quad (3)$$

Sinusoidal positional encodings [22] are then added:

$$\text{PE}_{(t,2k)} = \sin\left(\frac{t}{10000^{2k/d}}\right), \quad (4)$$

$$\text{PE}_{(t,2k+1)} = \cos\left(\frac{t}{10000^{2k/d}}\right). \quad (5)$$

4.3 Transformer Encoder with Macro Skip Connection

The encoder consists of L identical post-LayerNorm Transformer blocks, each applying multi-head self-attention (MHSA) followed by a position-wise feed-forward network (FFN) with GELU:

$$\mathbf{a}^{(\ell)} = \text{MHSA}\left(\mathbf{H}^{(\ell-1)}\right), \quad (6)$$

$$\mathbf{H}^{(\ell)} = \text{LayerNorm}\left(\mathbf{H}^{(\ell-1)} + \mathbf{a}^{(\ell)}\right), \quad (7)$$

$$\mathbf{H}^{(\ell)} = \text{LayerNorm}\left(\mathbf{H}^{(\ell)} + \text{FFN}\left(\mathbf{H}^{(\ell)}\right)\right). \quad (8)$$

The skip variant (**REnFormer-Skip**, used in the foundation-model comparison) additionally stores the projected, position-encoded input $\mathbf{H}^{(0)}$ and re-injects it after the final block:

$$\mathbf{H}^{\text{out}} = \text{LayerNorm}\left(\mathbf{H}^{(L)} + \mathbf{H}^{(0)}\right), \quad (9)$$

a macro residual that gives the head direct access to the normalised input pattern—useful for a strongly diurnal signal whose next day resembles a scaled copy of recent days. With $T = 168$ (7-day lookback) and $d = 128$, the $\mathcal{O}(T^2d)$ attention cost is negligible on a single GPU.

4.4 Probabilistic Forecast Head

The encoded representation of the last token, $\mathbf{h}_T^{(L)} \in \mathbb{R}^d$, is pooled as a summary of the full context. Two parallel linear layers project this to the forecast horizon:

$$\boldsymbol{\mu} = \mathbf{W}_\mu \mathbf{h}_T^{(L)} \in \mathbb{R}^H, \quad (10)$$

$$\log \boldsymbol{\sigma} = \mathbf{W}_\sigma \mathbf{h}_T^{(L)} \in \mathbb{R}^H, \quad (11)$$

$$\boldsymbol{\sigma} = \exp(\log \boldsymbol{\sigma}). \quad (12)$$

The log-space parametrisation guarantees $\sigma_h > 0$. During inference, the point forecast is $\hat{y}_h = \mu_h$ and a 90% prediction interval is $[\mu_h - 1.645\sigma_h, \mu_h + 1.645\sigma_h]$, all de-normalised to MW.

4.5 Training

REnFormer is trained end-to-end by minimising the masked NLL in Eq. (2) using Adam with learning

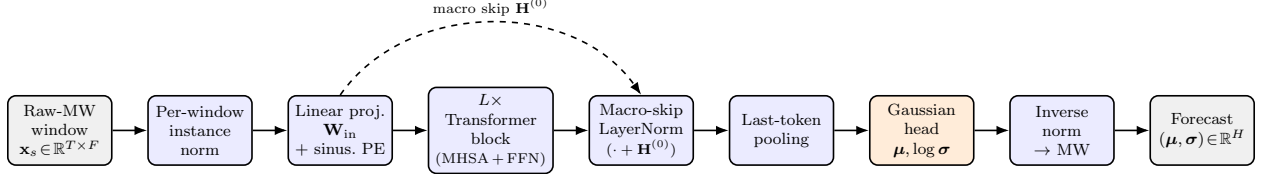


Figure 1: REnFormer architecture. A shared encoder processes each site’s lookback window independently. Each raw-MW window is renormalised by its own statistics, projected and position-encoded, passed through L post-LayerNorm Transformer blocks, and re-combined with the projected input via a macro skip connection (dashed). The last token is pooled and projected to Gaussian parameters for the forecast horizon, then de-normalised back to MW. Total: 536 k parameters.

rate 3×10^{-4} , cosine annealing, and a batch size of 64 windows. Supervision is in raw MW (the normalisation is inverted inside the forward pass), so the loss directly reflects operational error. Each window is drawn uniformly at random from all sites and dates; sites receive no explicit identifier, forcing the model to learn site-invariant temporal patterns. Table 1 lists the full configuration.

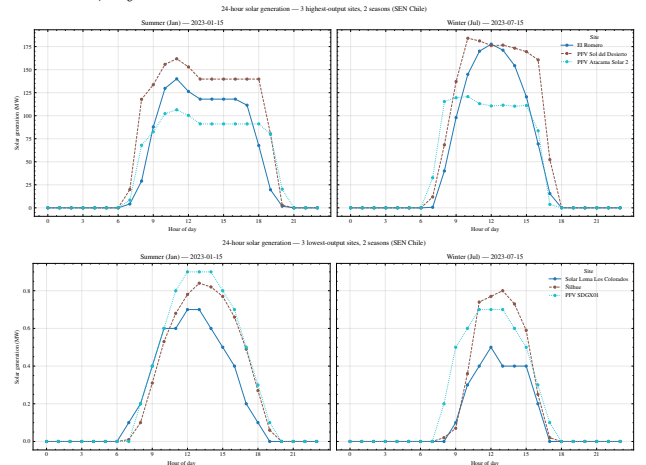
Table 1: REnFormer hyperparameters.

Hyperparameter	Value
Model dimension d	128
Attention heads A	4
Encoder layers L	4
FFN dimension	256
Dropout	0.1
Lookback T	168 h (7 days)
Horizon H	24 h
Batch size	64
Learning rate	3×10^{-4}
Epochs	50 ($\times 1000$ steps)
Intermittency threshold ϵ	0.1 MW
Total parameters	536,624

- **291 solar plants** spanning the Atacama Desert to the south of Chile, with capacities from < 1 MW to ~ 390 MW
- **47,400 hours** (5.4 years) of generation per site, ≈ 13.8 million site-hours in total
- **6.49 million** non-zero hourly observations
- Intermittent fraction: 57.9% of timesteps with generation ≤ 0.1 MW (nighttime + overcast)

We apply a chronological split: training on 2021–2023 (26,280 h), validation on 2024 (8,784 h), and test on 2025-01-01 to 2026-05-29 (12,336 h), yielding 7.59M / 2.50M / 3.53M sliding windows respectively. The test period is strictly future relative to training, so newer plants and fleet growth are part of the evaluation, matching deployment conditions.

Figure 2: Highest and lowest generation sites in the test set, by season.



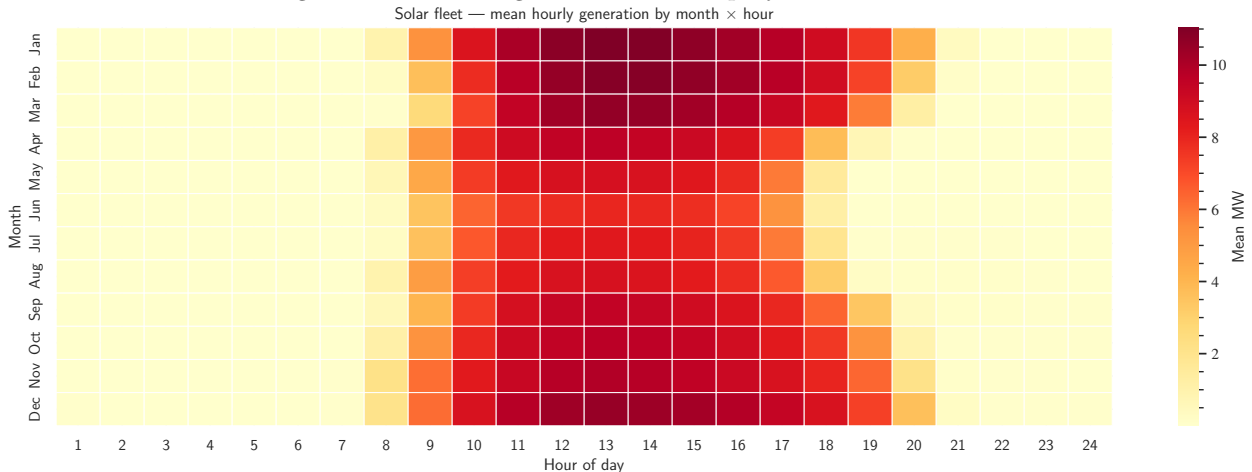
5 Experiments

5.1 Dataset: SEN Chile

We evaluate on the *Generación Real* dataset published by Chile’s National Energy Coordinator (CEN¹), covering the national electricity system (SEN) from 2021-01-01 to 2026-05-29. The raw export contains 817 grid-connected solar units; after removing sites with more than 10% missing hours, the corpus contains:

¹<https://www.coordinador.cl/operacion/graficos/operacion-real/generacion-real/>

Figure 3: Solar fleet generation heatmap by month and hour.



5.2 Comparison Protocol

The central comparison is REnFormer-Skip against **TimesFM 2.5** (200M-class checkpoint, 231,289,280 parameters) used *zero-shot*, the strongest available foundation-model comparator for this task [1, 5]. Both models receive identical inputs: the same 507 non-overlapping 168-hour raw-MW windows per site from the test period (stride 24 h), forecasting the next 24 hours. Neither model is given site identifiers or meteorological covariates. To characterise behaviour as the fleet grows, we sweep the number of sites $N \in \{10, 60, 110, 160, 210, 260, 291\}$ (sites in fixed order; 291 = full fleet).

5.3 Evaluation Metrics

For point forecasts we report mean absolute error (MAE) and root mean squared error (RMSE) in MW, computed only on non-trivial timesteps ($m_{s,t} = 1$, i.e., generation > 0.1 MW), to avoid inflating accuracy with trivially correct nighttime zeros. For probabilistic quality we report the *continuous ranked probability score* (CRPS), which jointly evaluates sharpness and calibration:

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbf{1}[z \geq y])^2 dz, \quad (13)$$

where F is the predicted CDF. We additionally report parameter count and wall-clock inference latency per 24-hour forecast (ms/forecast) on identical hardware.

5.4 Main Results

Table 2 reports the full-fleet (291 sites) comparison. REnFormer-Skip matches TimesFM 2.5 on MAE to

three decimal places while improving RMSE by 3.2% and CRPS by 5.1%—with $431\times$ fewer parameters and $\sim 640\times$ lower per-forecast latency. Forecasting the entire national fleet for the full 17-month test period takes 5.3s for REnFormer versus 57 minutes for TimesFM on the same GPU.

5.5 Scaling with Fleet Size

Table 3 sweeps the comparison from 10 to 291 sites. Three observations stand out. First, REnFormer’s *probabilistic* advantage is systematic: it achieves lower CRPS and RMSE than TimesFM at all seven fleet sizes (Fig. 4). Second, on MAE the two models are within noise of each other at every scale except the smallest subset, where REnFormer is 10.3% better—the first sites in the fleet ordering are well-represented in training and the specialist exploits them. Third, REnFormer’s per-forecast latency is flat (≈ 0.037 ms) across scales, whereas the foundation model’s ≈ 24 ms per forecast compounds to nearly an hour of GPU time at fleet scale.

5.6 Ablation Study

On the full sliding-window test set, the base variant and the skip variant are statistically indistinguishable (Table 4), indicating that at this model size the per-window renormalisation—not the specific encoder topology—carries most of the specialisation benefit. We adopt the skip variant for the foundation-model comparison as it was marginally stronger on the most recent held-out window.

Now focusing on the most recent 168-hour window (the last test window for each site), the skip variant’s macro residual gives it a clear edge over the base

Table 2: Full-fleet test-set comparison (291 sites, 507 non-overlapping windows per site, non-trivial steps only). MAE/RMSE/CRPS in MW.

Method	MAE	RMSE	CRPS	Params	ms/fc
TimesFM 2.5 (zero-shot)	1.936	6.541	1.450	231.3M	23.0
REnFormer-Skip	1.936	6.333	1.376	0.54M	0.036

Table 3: Scaling sweep over fleet size N . R-S = REnFormer-Skip, TFM = TimesFM 2.5 zero-shot. Best per row in **bold**.

N	MAE		RMSE		CRPS	
	R-S	TFM	R-S	TFM	R-S	TFM
10	0.560	0.625	0.897	1.066	0.392	0.470
60	1.037	1.040	4.413	4.435	0.734	0.782
110	0.819	0.839	3.292	3.323	0.579	0.633
160	1.827	1.830	6.531	6.686	1.299	1.368
210	2.287	2.281	7.148	7.383	1.626	1.705
260	2.023	2.027	6.535	6.760	1.439	1.517
291	1.936	1.936	6.333	6.541	1.376	1.450

Table 4: Architecture variants, full sliding-window test set (3.53M windows, 24h horizon).

Variant	MAE	RMSE	CRPS
REnFormer (Base)	1.846	6.174	1.322
REnFormer-Skip	1.851	6.175	1.323

variant, suggesting that direct access to the normalised input pattern is helpful for forecasting the next day. A regime where the diurnal pattern is strong and the model can exploit the resemblance between recent days and the next day. This is the variant we use for the foundation-model comparison, and the one we recommend for operational use.

Table 5: Architecture variants, Last 168-hour window 24-hour forecast.

Variant	MAE	RMSE	CRPS
REnFormer (Base)	3.3131	9.0442	2.3878
REnFormer-Skip	3.0899	8.1277	2.1354

6 Discussion

When does the specialist win? Our results delineate the regimes cleanly. With 5+ years of in-domain history, a 536 k-parameter specialist matches a 231M-parameter foundation model on point accuracy and

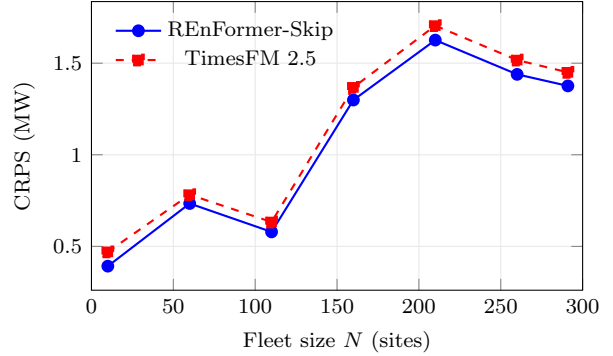


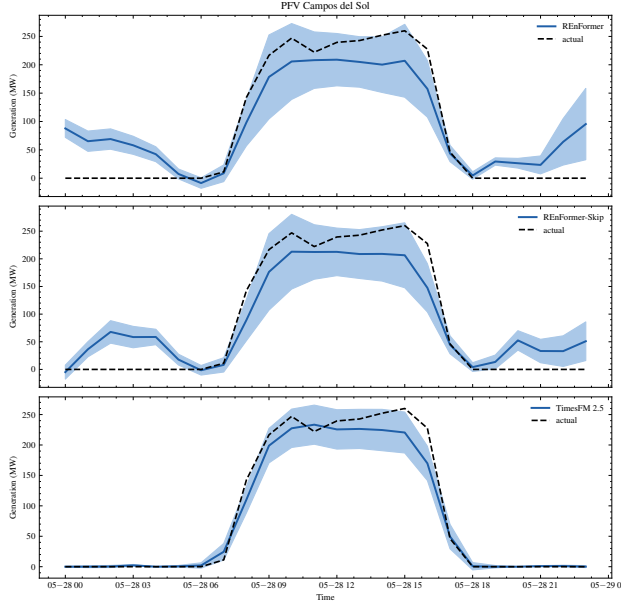
Figure 4: Probabilistic quality (CRPS, lower is better) versus fleet size. REnFormer-Skip (solid) sits below TimesFM 2.5 zero-shot (dashed) at every fleet size; the metric level tracks the mix of plant capacities added as N grows, not a monotone trend.

beats it on probabilistic quality—consistent with findings that the architecture around the forecaster, not pre-trained scale, drives accuracy [6], and with the success of compact models built with the right inductive biases [8,9]. The foundation model retains a genuine advantage where the specialist cannot go: plants with no history (cold-start) and grids without accumulated data. Lightweight adaptation of TSFMs [11] and few-shot compact foundation models [10] may eventually close the gap from the other side; a hybrid that backs off to a TSFM for young plants is a natural operational design.

Operational cost. For a system operator issuing hourly forecasts for hundreds of plants, the $\sim 640 \times$ latency gap is not cosmetic: it is the difference between forecasting the fleet in seconds on commodity hardware and dedicating GPU-hours per day, and it makes high-frequency re-forecasting and large ensemble runs feasible. The full REnFormer training run (50 epochs) completes on a single GPU, in line with the single-GPU training budgets reported for other small forecasters [7,9].

Why masking matters for solar. 57.9% of all site-hours are trivial (≤ 0.1 MW). A generic pre-trained model must spend capacity learning to reproduce zeros; the masked NLL (Eq. 2) lets a small model

Figure 5: Last-window 24 h forecast for one site from three models.



spend its entire 536 k-parameter budget on the operationally relevant daylight regime. We view this kind of domain-structure exploitation as the core reason small specialists stay competitive.

Limitations. REnFormer is trained without meteorological covariates (no NWP input), relying solely on generation history; both models in our comparison are therefore weather-blind, and adding irradiance forecasts would likely benefit both. The TimesFM comparison is zero-shot: fine-tuning the foundation model on SEN data would narrow the accuracy gap at the cost of its main selling point (no training) while leaving the $431\times$ parameter and inference gap intact. Relatedly, our comparison conflates two axes: REnFormer is an encoder-only model trained from scratch, whereas TimesFM is a decoder-only model pre-trained as a foundation model. The encoder-only design grants bidirectional attention over the full lookback and emits the 24h horizon in a single pass, avoiding the autoregressive error accumulation and sequential decoding of TimesFM; the decoder-only design, conversely, enables the large-scale next-token pre-training behind its zero-shot generalisation and flexible horizon. The architectural effect (bidirectional, direct-horizon vs. causal, autoregressive) and the training-regime effect (specialist vs. pre-trained) are thus entangled. Finally, our fleet-size sweep uses a fixed site ordering; metric levels across N reflect the mix of plant sizes added, not a pure scaling law.

7 Conclusion

We presented REnFormer, a 536 k-parameter domain-specific Transformer for probabilistic multi-site solar forecasting, and evaluated it like-for-like against the 231M-parameter TimesFM 2.5 foundation model on 291 plants of the Chilean national grid. The small specialist matches the foundation model’s point accuracy at full fleet scale, improves probabilistic skill at every fleet size evaluated, and runs $\sim 640\times$ faster per forecast. In a literature increasingly sceptical that scale is what makes time-series models work [6, 7, 9], our results add grid-scale, operationally grounded evidence: when in-domain history is abundant, a small Transformer with the right domain structure—per-window reversible normalisation and intermittency-masked likelihood—remains a competitive choice, with foundation models best reserved for cold-start and data-scarce regimes. Code and pre-trained weights will be released upon acceptance.

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